Modeling restricted set of values

x only takes values in a given set $\{a_1, a_2, \ldots, a_m\}$.

Model

Introduce m binary variables y_j, $j = 1, \ldots, m$ and the following constraints:

$$x = \sum_{j=1}^{m} a_j y_j,$$

$$\sum_{j=1}^{m} y_j = 1$$

$$y_j \in \{0, 1\}$$

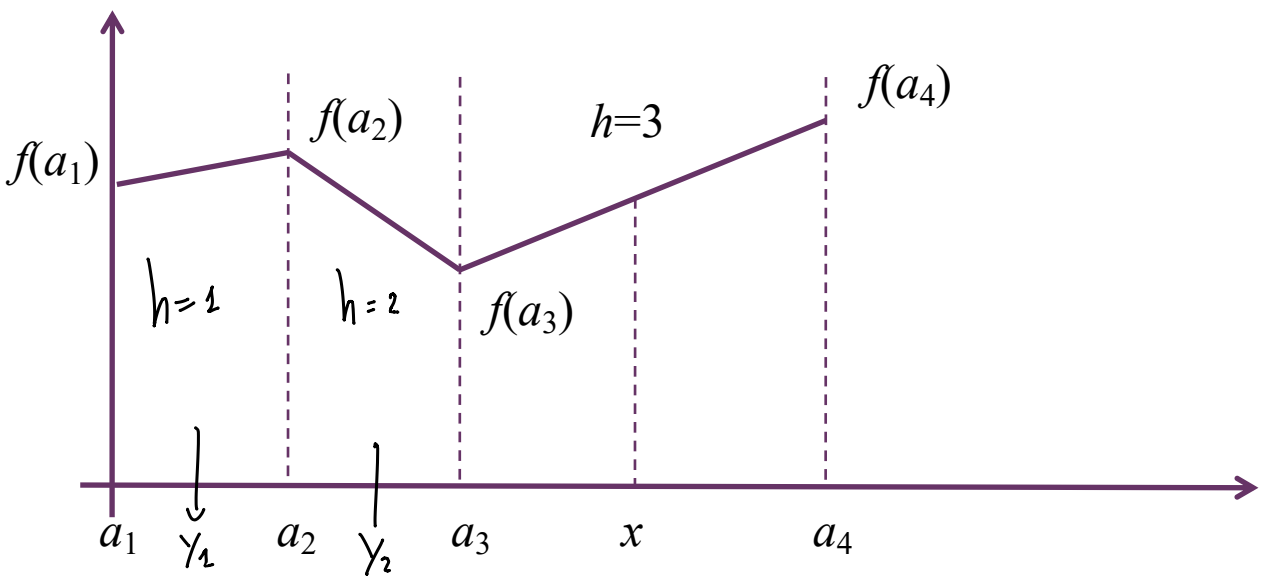
Piecewise linear cost function

A **piecewise linear** cost function is specified by m ordered pairs $(a_i, f(a_i))$

At a given point x in the h-th interval $[a_h, a_{h+1}]$ the function can be evaluated by:

$$f(x) = \lambda_h f(a_h) + \lambda_{h+1} f(a_{h+1})$$
$$\lambda_h + \lambda_{h+1} = 1$$
$$\lambda_h, \lambda_{h+1} \geq 0$$

il solver lo fa automaticamente



Piecewise linear cost function

Define **binary variables**

$$y_h = \begin{cases} 1 \text{ if } x \text{ is in the interval } [a_h, a_{h+1}] \\ 0 \text{ otherwise} \end{cases} \qquad \text{for } h = 1, \ldots, m-1$$

Define **continuous variables**

$$\lambda_h \geq 0 \text{ for } h = 1, \ldots, m$$

Objective function

$$\min \sum_{i=1}^{m} \lambda_i f(a_i)$$

Piecewise linear cost function

Constraints

$$\sum_{i=1}^{m} \lambda_i = 1,$$

soddisfa anche $\lambda_h + \lambda_{h+2} = 1$

posso stare solo in una partizione

$$\left.\begin{aligned} \lambda_1 &\le y_1, \\ \lambda_i &\le y_{i-1} + y_i, \\ \lambda_m &\le y_{m-1}, \end{aligned}\right\} \qquad i \in [2, \ldots, m-1],$$

$$\sum_{i=1}^{m-1} y_i = 1,$$

fanno in modo che solo λ_i e λ_{i+2} siano "in uso" ovvero quelli che contengono x

This set of constraints implies that there are only two "active" λ-s corresponding to interval in which x lies, i.e.,

$$y_j = 1 \text{ implies } \lambda_i = 0 \;\; \forall i \neq j, j+1.$$

Se vogliamo una funz. approssimata meglio, allora devo aumentare gli intervalli

Piecewise linear cost function

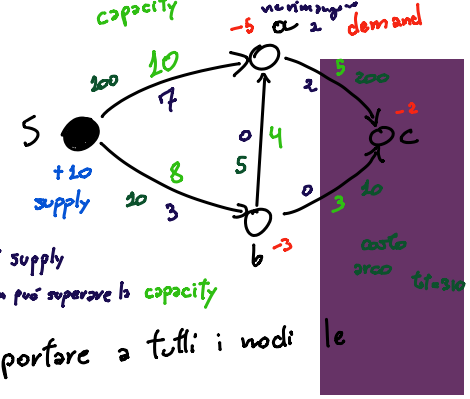
Formulation

$$
\begin{aligned}
\min \sum_{i=1}^{m} \lambda_i f(a_i) \\
\sum_{i=1}^{m} \lambda_i = 1, \\
\lambda_1 \leq y_1, \\
\lambda_i \leq y_{i-1} + y_i, \qquad i \in [2, \ldots, m-1], \\
\lambda_m \leq y_{m-1}, \\
\sum_{i=1}^{m-1} y_i = 1, \\
\lambda_h \geq 0 \text{ for } h = 1, \ldots, m \\
y_i = \{0, 1\} \text{ for } i = 1, ..., m-1
\end{aligned}
$$

Fixed Charge Network Flow Problem

$(s,b) \neq (b,s)$

inizio fine



quantità di supply spedita, non può superare la capacity

goal: portare a tutti i nodi le supply

direzionato

Given

A directed graph $G=(N,A)$ with a source node s

A demand vector

$$b \in \mathbb{Z}^{|N|} \text{ such that: } b_s > 0, b_i \leq 0 \; \forall i \in N, i \neq \{s\} \text{ and } \sum_{i \in N} b_i = 0$$

A capacity vector $u \in \mathbb{Z}^{|A|}$

A cost vector $c \in \mathbb{Z}^{|A|}$ representing the arcs activation cost

se non uso l'arco il costo = 0

Find

A feasible flow vector

$$0 \leq x \leq u$$

that minimizes the cost of activated arcs

Fixed Charge Network Flow Problem

Variables

Flow variables

supply shipped on an arc

arc capacity

$$0 \leq \overbrace{x_{ij}} \leq \overbrace{u_{ij}} \quad \forall (i,j) \in A$$

Activation variables

$\bar{c}$ on mixed problem

$$y_{ij} = \begin{cases} 1 \text{ if arc } (i,j) \text{ is activated,} \\ 0 \text{ otherwise.} \end{cases}$$

Objective function

$$\min \sum_{(i,j) \in A} c_{ij} y_{ij}$$

Fixed Charge Network Flow Problem

Constraints

Flow balance constraints

supply che entrano

supply che escono

supply richieste dal nodo i

$$\sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} = b_i \quad \forall i \in N$$

Activation constraints

$$0 \leq x_{ij} \leq \underbrace{u_{ij} y_{ij}}$$

in questo modo sono costretto a porre $y_i = 1$ se voglio usare l'arco

Fondamentale

altrimenti la soluzione sono tutti $y_i = 0$

Fixed Charge Network Flow Problem

Formulation

$$\min \sum_{(i,j)\in A} c_{ij} y_{ij}$$

$$\text{s.t.}$$

$$\begin{aligned}
\sum_{(i,j)\in\delta^+(i)} x_{ij} - \sum_{(j,i)\in\delta^-(i)} x_{ji} &= b_i \quad \forall i \in N \\
x_{ij} - u_{ij} y_{ij} &\leq 0 \quad \forall (i,j) \in A \\
x &\geq 0 \\
y &\in \{0,1\}^{|A|}
\end{aligned}$$